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RESEARCH ARTICLE

Reducing Hallucinations in Legal AI: A Retrieval Augmented Generation-Based Model for Accurate Legal Guidance

SHREYA GUPTA[1], VISHAL AGRAWAL[1], YASH NEGI[1], SATHYA KARUNANITHI[1], AND ARUNKUMAR BALAKRISHNAN[2]

[1]School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore 632014, India
[2]Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal 560064, India

Corresponding author: Sathya Karunanithi (sathya.k@vit.ac.in)

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ABSTRACT With the emergence of Large Language Models (LLMs) in the field of Generative Artificial Intelligence (AI), many industries have been significantly transformed, including the legal sector. These models are commonly used for tasks such as summarizing lengthy documents, answering questions, and analyzing large volumes of textual information, which helps improve efficiency in professional settings. However, LLMs are known to generate information that may appear convincing but is not supported by facts, a problem commonly referred to as hallucination. In fields like law, accuracy and reliability are critical. Common LLM errors can have serious consequences. To address this issue, this work proposes a Retrieval-Augmented Generation (RAG)-based approach that incorporates LLMs with a document retrieval mechanism. A custom vector database of 17 recent Indian legal documents sourced from the official Indian government legislation website is built using an embedding model. This database helps find parts of these documents that are relevant to what a user is asking. When the system uses the text it found to answer questions, it is less likely to give information or make up references. The study displays six ways of combining two types of models to see what works best. The combination of Llama2 and all-mpnet-base-v2 turned out to perform the best with a BERTScore F1 of 56.87%. This shows that using legal documents to help answer questions can make the system more reliable.

INDEX TERMS Generative AI, large language models (LLMs), hallucinations, legal AI, retrieval-augmented generation (RAG), AI ethics.

I. INTRODUCTION

The rapid advancement of language models has changed how natural language processing is done and a lot of other things. AI systems can now do complicated stuff. They can summarize what people say and answer questions in a very natural way. Large language models like GPT-4 are trained on large amounts of data. That lets them generate coherent, contextually relevant content. They have a lot of capability, but they also have a big problem. They can make things up. This is called hallucinations. The model invents information that sounds good but is wrong or completely made up. In fields like law, where accuracy is crucial this can cause serious trouble. It can lead to bad legal advice, incorrect references to cases or misinterpretation of laws. Precision and factual reliability matter in the real world. Errors in advice can have serious real-world implications, including financial penalties, damage to reputation or flawed judicial outcomes.

Even though AI is being used more and more in law there is still a gap in addressing the issue of hallucinations in legal content generated by large language models. While AI has a lot of potential to help professionals, the risk of inaccurate or fabricated information undermines the trust and reliability of AI systems in the legal sector. This is a challenge that needs

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to be addressed with more robust solutions that ensure legal AI systems can deliver precise, verified information.

Because lawyers are relying more and more on AI systems to make decisions, this work focuses on stopping hallucinations in AI. There have been cases where lawyers have cited non-existent court cases generated by AI, which shows the dangers of relying on unverified outputs. If legal responses are not accurate it can harm the credibility of AI. This also exposes legal practitioners to professional risks and their clients to significant legal harm. As AI becomes a part of legal workflows there is a clear need for solutions that can ensure the accuracy and reliability of AI-generated legal content.

One way to address this issue is by using Retrieval-Augmented Generation (RAG)-based approach. RAG models combine the power of language models with information retrieval systems, which allows AI to pull relevant verified data from trustworthy sources like legal databases, laws, and practice guides before generating responses. This makes the AI's outputs more accurate by basing them on time trustworthy legal information, which reduces the risk of hallucinations. By using this retrieval mechanism, RAG models offer a way to develop AI systems that deliver accurate, reliable, and legally sound guidance.

This work focuses on developing and evaluating a system based on RAG that is designed to reduce hallucinations in question-answering tasks. The study looks at how different combinations of embedding and LLMs affect the quality of the grounding and the accuracy of the responses when applied to documents. By focusing on retrieval-based contextualization the proposed framework aims to improve the reliability of AI-assisted information systems while minimizing outputs that are not supported or are completely made up.

II. RELATED WORKS

Recently, the rise of LLMs and generative AI has changed various areas, including significant advancements in areas such as legal question-answering systems, where the need for precision and knowledge is paramount. This section discusses the development of LLMs towards generative AI, the incorporation of RAG in improving these models, and the incorporation of Chain of Thought (CoT) prompting in improving these models in the area of law. These models have addressed the issues of information correctness and depth in the area of law. The most important works in this area have been compared and discussed in Table 1 below.

A. LLMs FOR COMPLEX LEGAL REASONING TASKS

There is an increasing focus on explicitly designing and assessing the reasoning of legal LLMs. A 2025 paper on Legal Reasoning in LLMs uses Chain of Thought traces as an interpretation tool and presents a benchmark and methodology for evaluating the quality of multi-step legal reasoning in LLMs, as measured by humans on aspects such as factual grounding and logical coherence [9]. Similarly, the LegalΔ framework proposes a reinforcement learning approach to train legal LLMs with dual modes of direct answers and reasoning-augmented answers with the use of information gain rewards to improve the accuracy of the answers as well as the faithfulness of the underlying reasoning'' [10]. These contributions represent a move away from legal LLMs as closed-box predictors to legal LLMs as systems with internal workings that can be optimized and interrogated. This is more directly aligned with the objectives of minimizing hallucinations and maximizing trustworthiness [11].

A culturally sensitive test framework for analyzing hallucinations in GPT models was proposed by McIntosh et al., emphasizing the importance of effective testing in sensitive application areas. An in-depth survey on RAG architectures was conducted by Ren et al., emphasizing the importance of external knowledge incorporation in LLMs for enhanced multilingual support and key trust factors [2]. Yang et al. surveyed the applications of LLMs in legal question-answering systems, noting the importance of model tuning and sequential data processing in enhancing the quality of reasoning in legal AI systems to automate complex tasks accurately [12]. Guan et al. proposed the LDMLSV model, which is based on SHAP explanations and soft voting, for the prediction of critical paths in labor dispute resolution, showing the potential for explainable multi-step decision processes to be represented by models for high-stakes decisions [3].

B. RETRIEVAL-AUGMENTED GENERATION (RAG)

RAG has been recognized as an important technique to improve the contextual appropriateness and factual correctness of LLM responses, especially for specific domains like law, where precise information retrieval is of utmost importance. LawRAG describes a retrieval-augmented system for judicial question answering on Australian case law, incorporating optimized vector embeddings with a large language model, along with the proposal of a new ''parent document'' retrieval method to take into consideration the hierarchical nature of judgments. [13]. A 2025 RAG-based question-answering framework for Korean laws and cases constructs a corpus of statutes and precedents, uses dense retrieval to fetch relevant provisions and decisions, and prompts an LLM to produce answers along with explicitly cited ''Relevant Laws'' and ''Related Cases,'' thereby directly targeting hallucination reduction through structured grounding [14]. In the Indian context, an Augmented Question-guided Retrieval (AQgR) approach for case law proposes using information about the ''question of law'' in addition to factual patterns to improve precedent retrieval, reflecting how real-world legal research mixes issue-based and fact-based search [15].

Fei et al. introduced InternLM-Law, an open-source model optimized for Chinese law that surpasses GPT-4 on multiple legal subtasks, showing the value of domain-specific training and retrieval for improving accuracy [4]. Nguyen et al. also worked on integrating legal practice guides with retrieval-based architectures to improve question-answering systems

TABLE 1. Summary of papers related to LLM and RAG models in legal and specialized domains.

Paper Title	Publication Year	Methodology	Results & Discussion	Limitations
LLM and RAG Architectures in IT Job Market [1]	2024	LLM and RAG architectures for information retrieval	Improved data scarcity handling, retrieval efficiency	LLM complexity, variability in RAG, open-source limitations
RAG Architectures with External Knowledge [2]	2024	Survey of RAG architecture integration with external knowledge	Enhanced LLM performance, trustworthiness, multilingual capabilities	Trustworthiness and external knowledge quality challenges
LDMLSV for Labor Dispute Resolution [3]	2024	SHAP and soft voting with Random Forest, Extra Trees, CatBoost	0.90 accuracy in predicting critical paths	Domain-specific limitations, need for further generalization
InternLM-Law for Chinese Legal Queries [4]	2024	Two-stage fine-tuned LLM trained on over 1 million legal queries	Outperformed GPT-4 on 13 of 20 legal subtasks	Focused on Chinese law, potential limits in global application
SaulLM-7B for Legal Text Processing [5]	2024	Mistral 7B architecture, fine-tuned on 30B tokens of legal text	State-of-the-art performance in legal text comprehension	Limited to English legal texts
Aalap for Indian Legal Tasks [6]	2024	Fine-tuned Mistral 7B for Indian law	Outperformed GPT-3.5-turbo in 31% of cases, matched GPT-4 in 34%	Legal reasoning over recall, domain-specific applicability
RAG-end2end for Open-Domain QA [7]	2023	RAG is adapted for specialized domains like healthcare	Enhanced ODQA, specialized knowledge integration	Challenges in domain-specific knowledge adaptation
SELF-RAG for Enhanced Factual Accuracy [8]	2023	Dynamic retrieval and self-reflection with "reflection tokens"	Improved factual accuracy and quality of information retrieval	Retrieving passages indiscriminately, reflection mechanism refinement

for complex legal cases, which illustrates how RAG can facilitate detailed context-specific answers to complex legal questions [16]. VM et al. offered practical advice on fine-tuning LLMs for enterprise use cases such as legal domains and offered advice on strategies and deployment considerations for RAGs that is relevant to the development of legal assistants [17].

These basic methods that helped find and use information are still really important. Zhao et al. came up with SELF-RAG, a framework that combines dynamic retrieval with self-reflection to improve factual accuracy [8], while Wang et al. introduced RAG-end2end for healthcare question-answering, showing that architectures which are retrieval-focused can be used in other high-stakes domains [7]. Fei et al. developed LawBench, which is a benchmarking dataset to evaluate LLMs on legal knowledge, mainly focusing on interpretability and factual alignment, reinforcing the importance of retrieval-based grounding in legal tasks [18]. Sovrano et al. contributed a dataset for legal question-answering focusing on private international law, proving the suitability of retrieval-based systems in complex sub-domains, which are usually narrow [19]. Earlier systems like LawGPT 1.0 have successfully demonstrated the practical potential of retrieval-backed assistants for routine legal document generation and question-answering.

Recent doctrinal and empirical work has evaluated RAG's impact on hallucinations and reliability in legal practice. A 2025 Harvard JOLT digest article characterizes retrieval-augmented architectures as a promising design pattern for legal work, arguing that grounding LLM outputs in curated legal databases can substantially reduce fabricated citations [20]. At the same time, independent audits of leading AI legal research tools report non-trivial rates of mis-citations and hallucinated cases, even in products that internally rely on retrieval [21]. Practitioner-oriented reports describing how law firms deploy RAG-based assistants over commercial databases emphasize the need for robust retrieval strategies, careful prompt design, and human oversight [22]. Together, these works motivate a more fine-grained, empirical analysis of RAG pipelines in legal settings-precisely the direction taken in this work, which systematically compares multiple embedding-LLM combinations on a curated Indian legal corpus using both lexical and semantic evaluation metrics.

C. CHAIN OF THOUGHT (CoT) PROMPTING FOR ENHANCED LEGAL REASONING

CoT prompting is a methodology to enhance the logical flow and reasoning level of LLM responses, especially important for intricate legal queries. The 2025 "Unpacking Legal Reasoning in LLMs" research proposes a benchmark to induce, examine, and rank CoT responses over a range of legal reasoning problems, where the quality of fact recognition, law application, and logical chaining is evaluated through human assessment [9]. This work positions CoT as a performance-enhancing prompt pattern and also as a tool for interpretable assessment, describing how LLMs reason about law. LegalΔ goes one step further by including CoT into the training objective via reinforcement learning with information-gain-guided rewards. This encourages models to generate reasoning steps that are both concise and informative in the long-run [10].

Prior research has identified the benefits of CoT in legal domains. Yang et al. surveyed various LLM-based legal question answering systems and concluded that "sequential data processing and reasoning can greatly help the automation of complex legal tasks [12]. Tiwari et al. introduced Aalap, which was a fine-tuned Mistral 7B model for Indian legal tasks. They proved that CoT prompting helps Aalap outperform GPT-3.5-turbo in 31% of test cases and match GPT-4 in 34%, highlighting CoT's role in contextually nuanced reasoning [6]. Sasidharan and Rahulnath came up with a way to find relationships in documents, based on graphs. It follows the principles of CoT by showing how things are connected to each other. This means it clearly shows the relationships between entities and provisions in documents [23]. Hoang Anh et al. analyzed CoT's meaningful impact on processing legal language, demonstrating that this approach effectively manages syntactic and semantic complexities specific to the legal domain. These are crucial for reliable AI-generated legal reasoning [24]. Nigam et al. evaluated LLMs on Indian legal question answering and found that CoT prompting allows models to deal with the complexity and cultural nuances of Indian laws in a more effective manner in terms of reasoning accuracy [25].

These developments suggest that future legal AI systems will combine retrieval-based grounding with explicitly optimized, interpretable reasoning processes. In this context, the RAG-based architecture and model-selection strategy pursued in this work can be viewed as complementary to CoT-oriented training and evaluation: retrieval reduces hallucinations by anchoring answers in authoritative legal texts, while CoT and related reinforcement-learning methods offer a path toward more transparent and controllable legal reasoning.

In contrast to LawRAG, Unpacking Legal Reasoning in LLMs, and LegalΔ, which each focus on a single axis of the problem—case-law–centric RAG design, CoT-based reasoning evaluation, or reinforcement-learning for CoT respectively, this work jointly optimizes retrieval and generation by systematically benchmarking six embedding–LLM combinations on a curated Indian statutory corpus using a diverse metric suite (ROUGE-1/2/L, BLEU, MPNET similarity, BERTScore F1, and MRR) over multiple queries. This not only to demonstrates hallucination reduction qualitatively but also quantitatively identifies the most effective pairing (all-mpnet-base-v2 + Llama2) and to show trade-offs against strong baselines like all-distilroberta-v1 + Mistral (7B) in a controlled, reproducible setting.

III. SYSTEM OVERVIEW

A structured methodology based on the use of information extraction, vector embeddings, and RAG is employed to resolve the problem of hallucinations in legal AI systems to ensure the accuracy of the responses generated. The system is based on the use of Indian legal resources such as ordinances and Regulations to make the responses generated more accurate. The system architecture is described in the figure below: Fig. 1 and its components of the architecture are described in the subsequent subsections.

A. INFORMATION EXTRACTION FROM RAW INDIAN LEGAL RESOURCES

The first step of the methodology is to obtain structured data from publicly available Indian legal resources, which are the foundational legal texts. The resources include legal documents like laws, amendments, ordinances, and regulations, which are usually available in an unstructured format like PDF files.

Since it is difficult to process an unstructured format, the obtained PDF files were converted into Markdown files with the extension .md to make it easier to process the text. The Markdown format was selected as it maintains the hierarchical structure of the text, i.e., it maintains the heading, list, and other structural elements of the text. This conversion is critical to avoid the possibility of hallucinations during the content generation process. The conversion is critical to ensure the accuracy of the model's output.

B. DATA CHUNKING

Once the information is extracted, the next step involves ''chunking'' the data. Legal documents, particularly constitutions and statutes, are typically lengthy and dense, making it impractical to process them as a whole. The extracted information is broken down into manageable ''chunks'' or smaller sections to make this easier to address. Chunking ensures that each segment of the text is concise and contextually self-contained. This allows the AI system to process smaller units of information at a time.

Every segment could relate to particular provisions of the Constitution, judicial decisions, or established legal precedents. This approach improves the accuracy of information extraction and enables the system to filter relevant sections of legal writings more effectively during the query handling process. Notably, chunking preserves the legal and logical coherence of the document, making certain that each segment offers distinct, legally sound information.

C. EMBEDDING THE DATA

After breaking down the legal data into pieces, the next step is to embed them, that is convert them into numbers to identify what the document or the legal data is actually about. The vectors represent legal language in a complicated manner, which can be easily processed by machine learning. In particular, a unique vector is created for a piece of legal data by using specific algorithms.

In this work, embeddings are created using pre-trained models that are fine-tuned on legal text. These embeddings allow the system to ''understand'' the meaning of each chunk and establish relationships between different legal concepts, cases, and statutes. The use of embeddings facilitates the accurate retrieval of relevant legal information later in the process.

D. CREATING A VECTOR DATABASE

Next, the vectorized representations of the legal chunks are stored in a specialized vector database. Unlike traditional databases that store data in tables, a vector database organizes and stores data based on its numerical vector representation. The vector database allows for fast and efficient similarity searches, where embeddings of queries can be compared against embeddings of legal texts to find the most relevant matches.

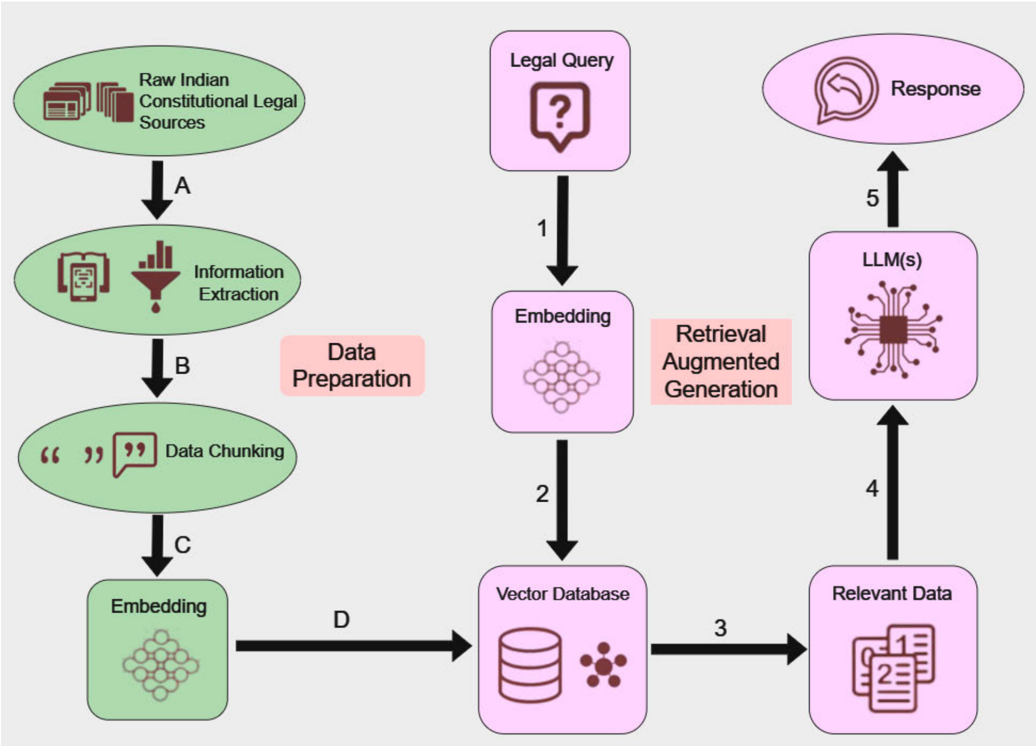

FIGURE 1. The architecture of the proposed model with all the sub-components.

This database serves as the core knowledge base for the AI system, housing the embedded representations of the Indian legal resources. When you look at this vector database the system can find the parts of the texts that are most important, to the users input. This happens fast even if the user asks the question in a different way, since the system can still identify the sections of the legal texts.

E. QUERY INPUT AND EMBEDDING

When you ask the system a question about laws like something about the constitution, court decisions or how to understand laws the system first tries to understand what you are asking. It does this by converting your question into a kind of code that captures the meaning of what you are asking. This way the system can compare your question with all the information it has even if you do not use the same words or phrases that are in the legal texts.

Converting the query into machine understandable code helps the system find the information for you from the huge amount of legal knowledge it has. By using this code, the system can tell how similar your question is, to the legal texts, based on what they mean not just if they have the same words. This process is called query embedding.

F. VECTOR SEARCH IN THE DATABASE

Once the query is converted to an embedding (a numerical format or machine-understandable code), it becomes easy for the system to compare it with the already existing data to find highest similarity matches and hence figure out the possible answer to your query. This is referred to as searching the vector database.

G. RELEVANT DATA EXTRACTION

When the search for vectors is done the system gets the legal information from the vector database. This information are the part that matches what the user is looking for. The system only takes the important information, so that the answer from the model is based on real and verified legal sources. This is a part of making sure that the model does not give hallucinated answers. The legal information that is taken out forms the base, for what the LLM say.

H. GENERATING THE FINAL RESPONSE USING AN LLM

Once the legal information is gathered it goes through a core LLM model to generate the final answer. This model takes the data and combines it into a clear and relevant response to the question. The system uses ways of generating answers making sure the response is complete, accurate and legally correct.

IV. DATASET CREATION

This work has employed a specific dataset to assess the performance of several embedding and core LLM model combinations based on the RAG-based technique.

The data set includes text-based legal ordinances or regulations from the recent past, retrieved from the official legislative website [26]. A total of 297 pages of legal documents were examined in this study, which covered various topics. Questions were formulated based on these documents (some with direct answers, some with indirect answers, and others with no answer in the dataset) and were tested using various models, followed by the scoring system for the assessment. The dataset for this study has been elaborated in Table 2 and formed the basis for the evaluation process. The provided Ordinances and Regulations were in PDF format and were converted into markdown format for better processing. The dataset was limited in size for this study as this work has been presented as a Proof of Concept (PoC). If the scope of the study were to be increased to include a larger dataset of legal documents, the computational needs for the embedding generation and the scoring process would have been higher and surpass the available resources, for the evaluation process using various model combinations.

TABLE 2. List of ordinances and regulations used for dataset creation.

S.No.	Title	Year
1	The Jammu and Kashmir Reorganisation (Amendment) Ordinance	2021
2	The Tribunals Reforms (Rationalisation and Conditions of Service) Ordinance	2021
3	The Insolvency and Bankruptcy Code (Amendment) Ordinance	2021
4	The Commission for Air Quality Management in National Capital Region and Adjoining Areas Ordinance	2021
5	The Indian Medicine Central Council (Amendment) Ordinance	2021
6	The Homoeopathy Central Council (Amendment) Ordinance	2021
7	The Essential Defence Services Ordinance	2021
8	The Narcotic Drugs and Psychotropic Substances (Amendment) Ordinance	2021
9	The Central Vigilance Commission (Amendment) Ordinance	2021
10	The Delhi Special Police Establishment (Amendment) Ordinance	2021
11	The Government of National Capital Territory of Delhi (Amendment) Ordinance	2023
12	The Lakshadweep Value Added Tax Regulation	2022
13	The Lakshadweep (Right to Public Services) Regulation	2022
14	The Lakshadweep Building Development Board (Repeal) Regulation	2022
15	The Lakshadweep Co-Operative Societies Regulation	2022
16	The Lakshadweep Panchayat Regulation	2022
17	The Lakshadweep Open Place (Prevention of Defacement) Regulation	2022

V. QUESTION ANSWERING PIPELINE

In the RAG setup for legal AI systems, the question-answering pipeline is a crucial part that seeks to utilize both the retrieved and the generative capabilities in order to offer accurate and relevant information for the user. The process for the question-answering pipeline in the RAG setup for legal AI systems involves the following: The process begins with the embedding model and the actual LLM. The embedding model seeks to convert the dataset and the user's query into higher-dimensional representations, referred to as embeddings, and stores them in the vector database. The core LLM then generates responses based on a prompt that integrates the retrieved context with the user's query.

A. EMBEDDING MODEL

The embedding model in this setup essentially fulfills the critical function of encoding text-based information into a format that recognizes the semantic meaning of the text. In the context of a legal AI system, this effectively enables the model to retrieve contextually relevant legal text from a database of legal documents when a query is entered by the user. This is achieved by effectively creating an embedding for both the dataset (stored as a .md file) and the query entered by the user, allowing for semantically similar legal text to be identified based on proximity in the embedding space.

The embedding models used in the current work are based on transformer models, which are used as the basis for natural language processing models today. The transformer models use self-attention and feed forward models to effectively understand the relationship between text and other words in a sentence. This is particularly important in legal texts because the meaning of the text is heavily dependent on contextual dependencies and the use of legal terms and phrases that are used to define the text and the clauses within the text, such as exceptions, conditions, definitions, and references to other sections within the text. The specific characteristics of the embedding model families used in this work are described, with emphasis on their applicability to the legal domain.

1) MPNet FAMILY

The MPNet models are really good at understanding what sentences mean. This is helpful for searches because people often ask questions in their own words using examples and simple language. The laws are written in formal language that is hard to understand. The MPNet model helps with this problem by looking at what the words mean not just if they are the same. This way it can find the laws even if the user's query does not use the same words as the law. The two models used in the work from this family include **All-mpnet-base-v2**, which works well for finding parts of laws, written in complicated language, that are relevant to what the user is looking for, and **Paraphrase-mpnet-base-v2** which is more appropriate for retrieval that emphasizes rephrasing, that is when users articulate legal scenarios in various manners (for instance, ‘‘quick resolution procedure’’ compared to ‘‘pre-pack insolvency procedure’’). It enhances the system's ability to retrieve information effectively with altered phrasing, ensuring it stays dependable across various types of queries and decreasing the chance of overlooking the accurate legal context because of differences in word choice.

2) MiniLM FAMILY

MiniLM contains small models that helps the system understand the meaning of the sentences it reads. They work well even if there are many documents to read. Legal systems have to answer several law-related questions and therefore need to look through many documents at the same time and this is where such models help. The two models used in the work from this family include **All-MiniLM-L12-v2**, which is very good at finding a balance between being correct and not requiring a lot of power. This is why it is good to use when you are looking at many queries in documents and want to analyze the relevance of each. **Paraphrase-MiniLM-L6-v2** on the other hand, is good for results and it works well when you need to search for something similar in meaning. When people talk about things they often use everyday language. The actual legal documents use big formal words which can be difficult to interpret. Paraphrase-MiniLM-L6-v2 is helpful because it can find what you are looking for when you use different words to describe it. This makes Paraphrase-MiniLM-L6-v2 a good choice when you need answers quickly and you want to know what to do in a situation right away.

3) DistilRoBERTa FAMILY

DistilRoBERTa is a version of RoBERTa. It helps the model understand sentences well while consuming limited memory. In a system that uses RAG for work DistilRoBERTa is very useful as it can understand legal sentences and parts of sentences pretty well. It does this without using much time or computer power. This makes DistilRoBERTa a choice for legal AI systems that need to find things quickly. The model used in this work from this family is **All-distilroberta-v1**, which acts as a general-purpose embedding model and performs well in capturing the semantic meaning of legal text such as definitions, statutory conditions, and procedural clauses. Although it may not be as semantically rich as larger embedding models, it provides stable retrieval performance for legal document chunks and can support efficient similarity search for grounding the LLM output in relevant statutory context.

These were chosen as embedding model options to support semantic retrieval from Indian legal documents in the proposed RAG pipeline. They were further tested in combination with different core LLMs to identify the best embedding-LLM combination for accurate and grounded legal question answering with reduced hallucinations.

B. CORE MODEL

The core model in a RAG-based legal AI system is responsible for generating the final response to the user query. After the embedding model retrieves relevant legal content, this content, along with the user query, is combined into a structured prompt. The prompt typically includes the query, the retrieved context, and instructions for generating a factually grounded response. The core model processes this prompt to produce a legally informed answer, reducing hallucinations and increasing the likelihood of providing accurate information. Overview of the core LLMs as used are given below.

1) Llama2

Llama2 is known for its accuracy and versatility in generating detailed, coherent responses. As a core model, it excels in the field of law by efficiently processing complex prompts. The performance of the model is characterized by generation of fluent language and relevant answers without hallucinations. The architecture of this model is based on the transformer architecture, composed of decoder-only layers. Each of the decoder-only layers has a multi-head self-attention mechanism, which enables the model to concentrate on the relevant parts of the prompt. In addition, the model has a feed-forward neural network component to process the embedding. The use of layer normalization is essential to ensure the efficient training of the model, use of rotary positional embeddings is essential to improve the performance of the model for long context sequences.

2) Llama 3.1 (8B)

Based on the Llama series, Llama 3.1 (8B) has been developed to offer more functionality in terms of understanding context and generating accurate output. This is because of its increased parameter set, which is now 8 billion. This allows for more detailed relationships to be made within data, which is then used to provide context-based answers. The model used to develop Llama3.1 is a transformer-based encoder-decoder model with multiple attention heads, which allows for parallel processing of text, enabling long-range dependency, as required.

3) Gemma2 (9B)

Gemma2 with 9 billion parameters can give detailed and smart answers. This model is also based on transformers but has 48 layers. These layers help Gemma2 understand texts by paying attention to many things at the same time and looking at the text in a special order. Gemma2 is very good at figuring out the relationships between parts of the text, even if they are far apart, which is really useful when looking at legal documents. This makes Gemma2 perfect for tasks that need a lot of thought and understanding.

The way Gemma2 is made is great for work that needs detailed and accurate answers based on a lot of information, which is exactly the requirement in legal systems. Moreover, it can handle big documents without getting confused, which means it can give high-quality answers even when the documents are very long, such as legal texts.

4) Qwen 2.5

Qwen 2.5 is recognized for its precision in handling factbased queries, which is essential in the legal field. This model is ideal for generating concise, factually accurate responses, especially in legal scenarios where specific information retrieval is prioritized over general text generation. Qwen 2.5 employs a transformer-based architecture with multiple optimizations for factual accuracy and contextual understanding. This helps minimize hallucinations by emphasizing fact-grounded response generation, thereby making it particularly suitable for applications requiring reliable legal interpretations.

5) Phi 3.5

Based on lightweight transformer models, Phi 3.5 efficiently combines performance with a parameter count of 3.8 billion, rendering it as a suitable option for efficiency with moderate computing resources. Its structure utilizes a transformer framework that is fine-tuned for efficient inference while still delivering solid performance. Phi 3.5 uses multi-head self-attention techniques to analyze various elements of the input text concurrently, improving its skill in recognizing complex relationships and contextual inter-dependencies. In addition, it incorporates rotary positional embeddings to manage sequential data effectively, enabling the model to excel in scenarios that require handling of sequential or multi-turn questions.

In summary, its design makes it suitable for reformulated legal queries and circumstances that need a balance of precision and quickness. These features establish Phi3. 5 as a practical choice for applications where computational performance is crucial, without losing the level of depth needed for sophisticated legal evaluation.

6) Mistral 7B

Mistral 7B is a balance of effectiveness and precision, delivering reliability in output for scenarios that demand in-depth responses with moderate processing requirements. The Mistral 7B framework is based on a transformer architecture with 7 billion parameters, with a focus on improving the balance between computation and expressiveness. Mistral makes use of a feature called multi-query attention (MQA), whereby a key-value pair is shared across multiple attention heads, thus minimizing memory usage while maintaining performance. The framework also employs RoPE, which is effective in encoding positional sequences and improving comprehension, even for long sequences of text.

The architecture of this model includes layer normalization and feed-forward networks (FFN), thus improving consistency and gradient flow throughout the training process. By utilizing dense activation and improving token management, Mistral 7B is effective in producing high-class embeddings and generating coherent text, thus making it a good fit for legal scenarios with RAG-based systems. The framework's effective utilization of parameters makes it applicable in scenarios where computation abilities are limited but at the same time makes accuracy and in-depth output crucial.

VI. EXPERIMENT AND RESULTS

To verify the performance of the RAG-based framework on the legal question-answering task, a series of experiments were performed. In the experiments, the performance of the proposed framework on the legal question-answering task was evaluated based on the ability to retrieve the relevant legal information, provide the correct responses, and tackle

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'''A company operating a small enterprise in Delhi faced financial difficulties
 due to the COVID-19 pandemic and is now considering filing for insolvency.
Can the company opt for a quicker resolution process to mitigate financial
damage and continue its operations? What legal framework applies to this situation?'''
```

FIGURE 2. A sample user query.

the challenging queries. Several evaluation metrics have been used to measure the performance of the proposed framework.

This section discusses more about how the experiments were set up and the kinds of questions were asked and finally how the RAG-based framework for question-answering was evaluated and which model worked best for for legal question-answering task.

A. SETUP

1) MODEL COMBINATIONS

The purpose of this study is to evaluate six combinations of embedding and core models in the framework of RAG-based approach for legal question answering. The framework was designed to answer complex legal questions by retrieving appropriate document passages while ensuring contextual accuracy. The various combination of models tested as a part of this work are as described below:

a. All-mpnet-base-v2 + Llama2: The embedding model all-mpnet-base-v2, which has been known to possess the ability to understand nuanced meanings, was integrated with Llama2 to form a joint entity to provide responses to queries.

b. All-mpnet-base-v2 + Llama3.1 (8B): Pairing all-mpnet-base-v2 with the Llama3.1 8B model gives more power to the model. This helps generate enhanced in-depth answers which are better for legal problems.

c. Paraphrase-mpnet-base-v2 + Gemma2 (9B): This combination used paraphrase-mpnet-base-v2 to manage rephrased legal text and Gemma2 for generating detailed responses, suited for queries with varied language patterns.

d. All-MiniLM-L12-v2 + Qwen 2.5: This lightweight embedding-core combination balanced speed and accuracy, aimed at providing rapid answers to straight-forward legal questions.

e. Paraphrase-MiniLM-L6-v2 + Phi3.5 (3.8B): The efficient paraphrase-MiniLM-L6-v2 was paired with Phi3.5 to handle questions which were paraphrased, particularly for environments with lower computational resources.

f. All-distilroberta-v1 + Mistral (7B): Combining the lightweight distilroberta embeddings with Mistral (7B), this setup focused on efficiently answering simple queries while maintaining contextual relevance.

B. QUERY CATEGORIES AND PROMPT STRUCTURE

The normal functioning of the suggested framework is such that a query is given to the model and the most appropriate answer is received from it. A sample query is given in Fig. 2 and an answer recieved from the model is given in Fig. 3. To test and measure the metrics, a sample reference answer is sent along with the query to test the answer given by the LLM

```
Based on the provided context information, it appears that the company is facing financial difficulties due
to the COVID-19 pandemic and is considering filing for insolvency. In this scenario, the company may be
able to opt for a quicker resolution process under Indian law.

In India, there are two main legal frameworks that govern corporate insolvency and bankruptcy: the
Insolvency and Bankruptcy Code, 2016 (IBC) and the Companies Act, 2013. Under IBC, a company can opt for a
quicker resolution process called the "fast track" process, which is designed to resolve insolvency cases
more quickly than the regular insolvency process.

To be eligible for the fast track process under IBC, the company must meet certain criteria, including:

1. The company must have a turnover of up to 100 crore rupees per annum.
2. The company must have a debt of up to 50 crore rupees per annum.
3. The company must not have been involved in any fraudulent or malicious activities.
4. The company must not be involved in any litigation or arbitration proceedings.

If the company meets these criteria, it can file a petition with the National Company Law Tribunal (NCLT)
under IBC to initiate the fast track process. Under this process, the NCLT will appoint a resolution
professional who will work towards resolving the insolvency within 100 days.

In addition to IBC, the Companies Act, 2013 also provides for a quicker resolution process called the
"voluntary winding up" process. This process is available to companies that are unable to pay their debts
and are willing to wind up their operations voluntarily. Under this process, the company can file a
petition with the NCLT to initiate the winding up process, which can be completed within 60 days.

It's important to note that the legal framework applicable to the company's situation may depend on various
factors, including the nature of its business, the amount and nature of its debts, and the specific
circumstances of its financial difficulties. Therefore, it is recommended that the company seek legal
advice from a qualified attorney before proceeding with any insolvency or bankruptcy process.
```

FIGURE 3. **Answer received from the best model.**

```
'''Yes, the company can opt for a pre-packaged insolvency resolution process
under the Insolvency and Bankruptcy Code (Amendment) Ordinance, 2021. Given
that the business qualifies as a micro, small, or medium enterprise (MSME),
the company can benefit from this efficient insolvency process designed to
facilitate quicker, cost-effective resolutions while preserving jobs and
maintaining business continuity.
Key provisions applicable:
Pre-packaged insolvency resolution process (Chapter III-A of the Insolvency
and Bankruptcy Code, 2016): This process allows MSMEs to resolve insolvency
without full-fledged insolvency proceedings, minimizing disruption.
Conditions: The company must not have undergone insolvency in the past three
years, and it needs the approval of at least 66% of financial creditors.
Procedure: The company must submit a base resolution plan, and upon approval,
the insolvency process will be completed within 120 days from initiation
(2023021047). This legal provision is tailored to support MSMEs impacted by
the pandemic, ensuring a smooth recovery.''',
```

FIGURE 4. **The reference answer for the given query.**

against the query. The reference answer is given in Fig. 4. Multiple such reference answers were sent and cross-checked for the models under test.

1) QUERY SELECTION

To assess model performance comprehensively, queries were categorized as follows:

a: DIRECT ANSWERS AVAILABLE

Some questions were designed to have exact matches within the database, enabling direct responses. This tested each model's ability to locate and accurately use direct legal references.

b: INDIRECT ANSWERS

Certain queries required interpretative answers based on the context rather than exact matches. These evaluated the models' capacity to derive answers indirectly from related legal content.

c: NO AVAILABLE ANSWERS

There were some questions that were not given corresponding answers in the database. This was to test how the models handled situations where there was a lack of information and to ensure that the models did not produce hallucinated responses.

There were reference answers to each question to be used as a reference when evaluating the models. This allowed for a comprehensive evaluation of the models' ability to produce legal and accurate responses to different types of questions.

2) MODIFIED PROMPT FOR LLM

The embedding model produced vector representations for the query, which were subsequently stored in the vector database. When the user query was submitted, vector representations for the query were produced and utilized for the retrieval of relevant context text from the database. The retrieved context text, the query, and the accompanying instructions were utilized for the generation of the modified prompt that functioned as the prompt for the generation of the response to the user's query. The prompt template consisted of:

a: USER QUERY

This was the direct question submitted by the user.

b: RETRIEVED CONTEXT

This section contains the top passages from the database that closely matched the query based on semantic similarity.

c: INSTRUCTION

Then a final guide was provided to the core model on handling cases with minimal context similarity or where information was limited for better results.

This structured prompt helped ensure that responses were grounded in relevant legal text while managing scenarios with varying context availability.

C. EVALUATION METRICS

The following metrics were used to evaluate each model's performance:

1) ROUGE (RECALL-ORIENTED UNDERSTUDY FOR GISTING EVALUATION)

ROUGE measures the n-gram overlap between generated responses and reference answers. For legal text, ROUGE-1 (unigram), ROUGE-2 (bigram), and ROUGE-L (longest common subsequence) were used, calculated as follows:

a: ROUGE-1

This measures unigram overlap. It was calculated by using (1).

$$\text{ROUGE-1} = \frac{\sum_{\text{unigrams}\in\text{Reference}\cap\text{Generated}} \text{Count(unigrams)}}{\sum_{\text{unigrams}\in\text{Reference}} \text{Count(unigrams)}} \tag{1}$$

b: ROUGE-2

This measures bigram overlap. It was calculated by using (2).

$$\text{ROUGE-2} = \frac{\sum_{\text{bigrams}\in\text{Reference}\cap\text{Generated}} \text{Count(bigrams)}}{\sum_{\text{bigrams}\in\text{Reference}} \text{Count(bigrams)}} \tag{2}$$

c: ROUGE-L

This measures the longest common subsequence overlap. It was calculated by using (3).

$$\text{ROUGE-L} = \frac{\text{LCS(Reference, Generated)}}{\text{Length of Reference}} \tag{3}$$

where LCS is the longest common subsequence.

2) BLEU (BILINGUAL EVALUATION UNDERSTUDY)

BLEU assesses response accuracy by comparing n-grams. It is calculated using a geometric mean of precision scores with a brevity penalty for short responses. For this experiment, BLEU focused on 1–4-gram overlaps, It was calculated by using (4)

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right). \tag{4}$$

where BP is the brevity penalty, w_n are weights (usually uniform), and p_n is the precision for n-grams.

3) MPNET SCORE

MPNET (Masked and Permuted Network) is an advanced embedding metric that enhances semantic similarity detection by building richer contextual embeddings. It improves on traditional masked language models by introducing token permutations during training. In MPNET, similarity is measured as the cosine similarity between embeddings of the generated and reference texts. It is depicted in (5)

$$\text{MPNET Similarity} = \frac{\sum_{i=1}^{n} A_i \cdot B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}}. \tag{5}$$

where A and B are embedding vectors. This score is particularly useful in legal contexts for its high alignment with semantic meaning.

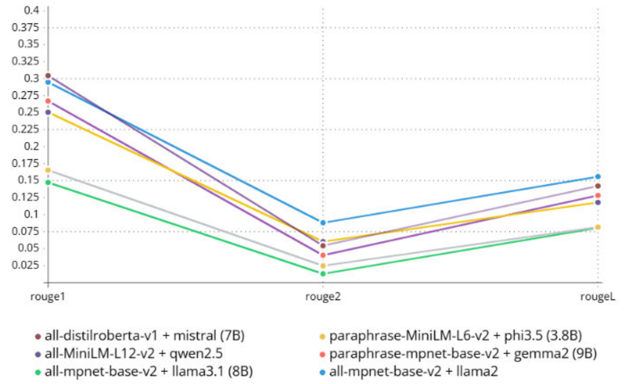

FIGURE 5. Graph plotting Rouge scores of different model combinations.

4) BERTScore F1

BERTScore F1 leverages BERT embeddings to measure semantic overlap by aligning token embeddings from generated and reference texts. The BERTScore F1 balances precision and recall based on cosine similarity between tokens. It is calculated as shown in (6)

$$\text{BERTScore F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

where Precision is the match of generated tokens to reference tokens, and Recall is vice versa. High BERTScore F1 values indicate strong alignment with the reference, ideal for ensuring accuracy in RAG-based legal responses.

5) MEAN RECIPROCAL RANK (MRR)

MRR measures the rank of the first correct answer returned by the model, with values close to 1 indicating high retrieval effectiveness. This is calculated using (7)

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}. \tag{7}$$

where $rank_i$ is the position of the correct answer for the ith query.

D. ANALYSIS

In order to test the efficiency of these models and compare them, they were tested by passing them through the aforementioned evaluation metrics. For the purpose of model evaluation, the ideal expected answer was passed through the evaluation functions, along with the answer obtained by the LLMs after processing the user queries. The evaluation functions provided a score for each metric, covering all six test questions used as user query inputs. An average score was obtained for each metric, giving a comprehensive idea about the relative efficiency of the models. The average score of the models from this evaluation process was used to analyze and compare the models' efficiency. The score, indicating the efficiency of the models on the evaluation criteria, is as follows in Table 3.

TABLE 3. Average scores of multiple questions comparison of models.

Models Combination	Rouge1	Rouge2	RougeL	BLEU	mpnet	BERT F1	MRR
all-mpnet-base-v2 + llama2	0.2949	0.08804	0.1559	15.4876	0.6846	0.5687	1
all-mpnet-base-v2 + llama3.1 (8B)	0.1473	0.01296	0.0805	2.637	0.5989	0.474	1
paraphrase-mpnet-base-v2 + gemma2 (9B)	0.2671	0.0402	0.1281	6.6455	0.6232	0.495	1
all-MiniLM-L12-v2 + qwen2.5	0.2506	0.0603	0.1179	7.0142	0.6723	0.5182	1
paraphrase-MiniLM-L6-v2 + phi3.5 (3.8B)	0.1654	0.0247	0.0816	4.4504	0.5725	0.4703	1
all-distilroberta-v1 + mistral (7B)	0.3044	0.0542	0.1421	11.7939	0.6578	0.5259	1

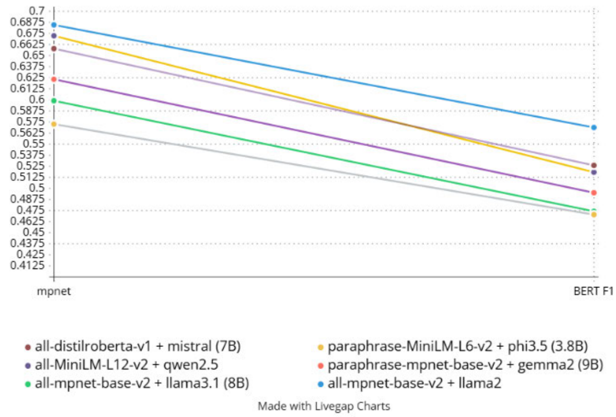

FIGURE 6. Graph plotting Sematic scores - mpnet and BERT F1 of different model combinations.

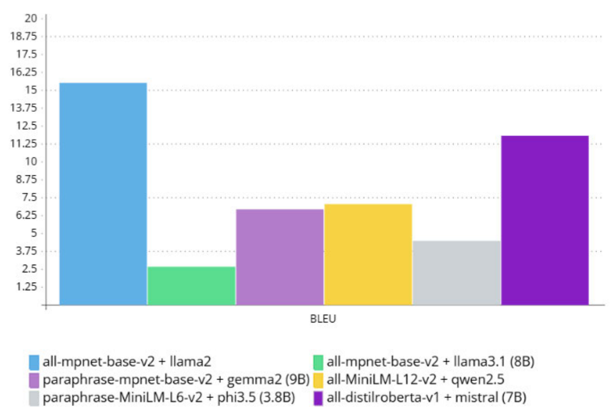

FIGURE 7. Graph plotting BLEU scores.

Fig. 5 shows the visual representation of the comparison of the ROUGE scores of all the model combinations. Among all the model combinations, the performance of the all-distilroberta-v1 with Mistral (7B) is the best in the case of the ROUGE-1 score. On the other hand, the performance of the all-mpnet-base-v2 with Llama2 is the best in the case of the ROUGE-2 and ROUGE-L scores.

Fig. 6 highlights the semantic evaluation metrics, such as MPNET similarity and BERT F1 scores, where the all-mpnet-base-v2 with Llama2 combination again stands out. These metrics emphasize the model's capacity to produce embeddings that align closely with the reference answers, demonstrating its strong semantic understanding and contextual relevance.

Fig. 7 shows a focus on the results related to the BLEU scores, where the precision of the n-gram results is compared across all the model combinations. Again, the all-mpnet-base-v2 with Llama2 model combination has proven to be the most effective. These visual results have collectively proven the superiority of the all-mpnet-base-v2 with Llama2 model combination in the generation of high-quality results in RAG-based legal AI applications.

These results show that the model combinations worked better in legal scenarios where accuracy and detail were important. Among all the model combinations, the results showed that the combination of all-mpnet-base-v2 + Llama2 produced the best results, with a ROUGE-1 score of 0.2949 and a BLEU score of 15.4876. This model combination showed the best semantic accuracy and was able to provide detailed responses to queries.

The combination of all-distilroberta-v1 + Mistral (7B) produced a balanced performance with ROUGE-1 of 0.3044 and BLEU of 11.7939. The combination excelled at answering straightforward legal questions while at the same time providing computational efficiency and speed. The combination is thus useful for cases where the generation of responses is critical. The configurations of all-mpnet-base-v2 + Llama3.1 (8B) and paraphrase-mpnet-base-v2 + Gemma2 (9B) showed high semantic accuracy with regard to the answering of paraphrased questions. The effectiveness of the combination was evident by the achievement of 0.5989 by MPNET F1 and 0.6232 by BERT F1. The combination is useful for cases involving rephrased or nuanced legal text. The combination of paraphrase-MiniLM-L6-v2 + Phi3.5 (3.8B) showed relatively lower performance with a BLEU score of 4.4504. The combination is thus limited with regard to the answering of complex legal text.

The results show that it is really important to choose the embedding-core model combinations carefully for the task. Llama2 and Mistral are choices for many different legal questions. Other combinations can also work well. It is necessary to pick models that can handle the complexity of the information that is put in and the kind of answer that is wanted. The right embedding-core model combinations are crucial for AI applications because they need to be able to handle the specific requirements of the task. Llama2 and Mistral are good at this. The key is to find the right combination, for the specific task. This is what makes legal AI applications work well.

VII. CONCLUSION

This study examines the use of a RAG approach to reduce hallucinations in legal AI systems. The results show that incorporating retrieved legal context helps improve the factual accuracy of generated responses, which is critical in domains such as law where incorrect information can lead to serious consequences. Among the configurations evaluated, the combination of all-mpnet-base-v2 as the embedding model and Llama2 as the core model achieved the best performance, with higher BERTScore F1 and BLEU scores across multiple queries. This indicates that appropriate model selection plays an important role in handling the complexity of legal language.

The findings also highlight the importance of grounding AI outputs in verified legal sources rather than relying solely on generative capabilities. This approach reduces the risk of producing plausible but incorrect information and makes such systems more suitable for professional use. In addition to performance improvements, the use of retrieval mechanisms supports better transparency and reliability, which are essential considerations in high-stakes applications.

Although the study focuses on Indian legal documents, the approach can be extended to other domains such as healthcare and finance. Overall, the results suggest that RAG-based systems provide a practical direction for improving the reliability of AI-assisted decision-making in structured knowledge domains.

VIII. LIMITATIONS AND FUTURE SCOPE

The current study is based on a relatively small set of legal documents due to computational constraints. Expanding the dataset to include a larger collection of statutory and regulatory texts is expected to improve system performance across a wider range of queries. Future work can also incorporate additional document types such as court judgments, legal agreements, and multilingual content to make the system applicable in more diverse legal settings.

The approach can be extended to other jurisdictions, including the US, UK, and EU, enabling broader applicability across different legal systems. Further fine-tuning and training may allow the model to handle more complex queries and larger datasets more effectively. The study also shows that while certain model combinations perform well overall, the system remains flexible and can be adapted based on specific requirements such as speed or semantic understanding.

Additional improvements may include optimizing the retrieval pipeline for faster performance, exploring more efficient search methods, and extending the system to handle multimodal data such as charts or structured legal information. Continuous updates using new legal documents can also help maintain the relevance of the system over time.

REFERENCES

[1] A. Mrinali and P. Gupta, "Enhanced cbir based secure cloth recommendation system using hybrid algorithm with explainable AI and rag descriptors," 2024. [Online]. Available: https://ssrn.com/abstract=4976203

[2] W. Fan, Y. Ding, L. Ning, S. Wang, H. Li, D. Yin, T.-S. Chua, and Q. Li, "A survey on RAG meeting LLMs: Towards retrieval-augmented large language models," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2024, pp. 6491–6501.

[3] J. Guan, Z. Yu, Y. Liao, R. Tang, M. Duan, and G. Han, "Predicting critical path of labor dispute resolution in legal domain by machine learning models based on Shapley additive exPlanations and soft voting strategy," *Mathematics*, vol. 12, no. 2, p. 272, Jan. 2024.

[4] Z. Fei, S. Zhang, X. Shen, D. Zhu, X. Wang, M. Cao, F. Zhou, Y. Li, W. Zhang, D. Lin, K. Chen, and J. Ge, "InternLM-law: An open source Chinese legal large language model," 2024, *arXiv:2406.14887*.

[5] P. Colombo, T. Pessoa Pires, M. Boudiaf, D. Culver, R. Melo, C. Corro, A. F. T. Martins, F. Esposito, V. Lúcia Raposo, S. Morgado, and M. Desa, "SaulLM-7B: A pioneering large language model for law," 2024, *arXiv:2403.03883*.

[6] A. Tiwari, P. Kalamkar, A. Banerjee, S. Karn, V. Hemachandran, and S. Gupta, "Aalap: AI assistant for legal & paralegal functions in India," 2024, *arXiv:2402.01758*.

[7] S. Siriwardhana, R. Weerasekera, E. Wen, T. Kaluarachchi, R. Rana, and S. Nanayakkara, "Improving the domain adaptation of retrieval augmented generation (RAG) models for open domain question answering," *Trans. Assoc. Comput. Linguistics*, vol. 11, pp. 1–17, Jan. 2023.

[8] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," 2023, *arXiv:2310.11511*.

[9] Y. Y. Chu, S.-C. Huang, and H.-L. Shao, "Unpacking legal reasoning in llms: Chain-of-thought as a key to human-machine alignment in essay-based nlu tasks," in *Proc. 5th Workshop Natural Log. Meets Mach. Learn. (NALOMA)*, 2025, p. 7.

[10] X. Dai, B. Xu, Z. Liu, Y. Yan, H. Xie, X. Yi, S. Wang, and G. Yu, "Legalδ: Enhancing legal reasoning in LLMs via reinforcement learning with chain-of-thought guided information gain," 2025, *arXiv:2508.12281*.

[11] T. R. McIntosh, T. Liu, T. Susnjak, P. Watters, A. Ng, and M. N. Halgamuge, "A culturally sensitive test to evaluate nuanced GPT hallucination," *IEEE Trans. Artif. Intell.*, vol. 5, no. 6, pp. 2739–2751, Jun. 2024.

[12] X. Yang, Z. Wang, Q. Wang, K. Wei, K. Zhang, and J. Shi, "Large language models for automated Q&A involving legal documents: A survey on algorithms, frameworks and applications," *Int. J. Web Inf. Syst.*, vol. 20, no. 4, pp. 413–435, Jul. 2024.

[13] L. K. Suresh Kumar, M. Yaseen, M. Junaid Adil, and V. Ramesh, "Lawrag: Retrieval-augmented generation for judicial question answering," *Propuls. Technol. J.*, vol. 46, no. 3, pp. 472–481, 2025. [Online]. Available: https://www.propulsiontechjournal.com/index.php/journal/article/view/9711/5894

[14] Y. Yu, Y. Y. Chu, S.-C. Huang, and H.-L. Shao, "RAG based question answering of Korean laws and precedents," in *Proc. 8th Fact Extraction Verification Workshop (FEVER)*, 2025, pp. 91–101.

[15] R. Gupta, M. Sharma, and P. Rao, "Augmented question-guided retrieval of Indian case law," 2025, *arXiv:2508.04710*.

[16] C. Doyle and A. D. Tucker, "If you give an LLM a legal practice guide," in *Proc. Symp. Comput. Sci. Law (CSLAW)*. New York, NY, USA: Association for Computing Machinery, 2025, pp. 194–205. [Online]. Available: https://doi.org/10.1145/3709025.3712220

[17] M. J. Raj, V. M. Kushala, H. Warrier, and Y. Gupta, "Fine tuning LLM for enterprise: Practical guidelines and recommendations," 2024, *arXiv:2404.10779*.

[18] Z. Fei, X. Shen, D. Zhu, F. Zhou, Z. Han, S. Zhang, K. Chen, Z. Shen, and J. Ge, "LawBench: Benchmarking legal knowledge of large language models," 2023, *arXiv:2309.16289*.

[19] F. Sovrano, M. Palmirani, B. Distefano, S. Sapienza, and F. Vitali, "A dataset for evaluating legal question answering on private international law," in *Proc. 18th Int. Conf. Artif. Intell. Law*, Jun. 2021, pp. 230–234.

[20] P. Johnston, "Retrieval-augmented generation (RAG): Towards a promising LLM architecture for legal work," *Harvard J. Law & Technol. Dig.*, Apr. 2025. [Online]. Available: https://jolt.law.harvard.edu/digest/retrieval-augmented-generation-rag-towards-a-promising-llm-architecture-for-legal-work

[21] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, and D. E. Ho, "Hallucination-free? Assessing the reliability of leading AI legal research tools," 2024, *arXiv:2405.20362*.

[22] "RAG for law firms: Automating legal research without hallucinations," *Alternates AI Blog*, 2025. [Online]. Available: https://www.alternates.ai/blog/ rag-for-law-firms-automating-legal-research-without-hallucinations

[23] A. K. Sasidharan and R. Rahulnath, "Structured approach for relation extraction in legal documents," in *Proc. 4th IEEE Global Conf. Advancement Technol. (GCAT)*, Oct. 2023, pp. 1–6.

[24] D. H. Anh, D.-T. Do, V. Tran, and N. L. Minh, "The impact of large language modeling on natural language processing in legal texts: A comprehensive survey," in *Proc. 15th Int. Conf. Knowl. Syst. Eng. (KSE)*, Oct. 2023, pp. 1–7.

[25] S. K. Nigam, S. K. Mishra, A. K. Mishra, N. Shallum, and A. Bhattacharya, "Legal question-answering in the Indian context: Efficacy, challenges, and potential of modern AI models," 2023, *arXiv:2309.14735*.

[26] (2024). *Indian Goverment Offical Site*. [Online]. Available: https://www.legislative.gov.in/legislative-references

SHREYA GUPTA is currently pursuing the B.Tech. degree in computer science and engineering with Vellore Institute of Technology, Vellore. With an outstanding academic record, she has a strong inclination toward research and development in artificial intelligence, machine learning, and data science. She has gained hands-on experience through prestigious internships and has contributed to projects involving generative models, NLP-backed prompt generation, and cyber defense strategies. She is proficient in deep learning and data visualization techniques. Her work on these projects earned her accolades, including the Best Paper Award at an IEEE conference. She is skilled in various programming languages and frameworks. She has been actively involved in various student leadership roles. She has organized, and participated in several technical and socio-cultural events,

VISHAL AGRAWAL is currently pursuing the B.Tech. degree in computer science and engineering with a specialization in bioinformatics with Vellore Institute of Technology, Vellore. With a stellar academic record, he has showcased exceptional skills in artificial intelligence, machine learning, full-stack development, and DevOps. He has interned at prestigious organizations where he contributed to building and developing solutions. He has excelled in competitive platforms and served as an active leader. He has received multiple merit scholarships and awards for academic and extracurricular excellence. His technical proficiency spans programming languages and tools.

YASH NEGI is currently pursuing the B.Tech. degree in computer science and engineering (CSE) specializing in bioinformatics with Vellore Institute of Technology, Vellore, Tamil Nadu, India. With a strong foundation in computer science and emerging technologies, he has demonstrated an aptitude for research and development in artificial intelligence, machine learning, data analytics, and web development. He has actively contributed to various academic and industrial projects, excelling in full-stack development, data analysis, and network security. He has gained experience guiding team projects, optimizing workflows, and exploring innovative solutions. He has attended numerous workshops, technical talks, and training programs, enhancing his expertise in cutting-edge technologies.

SATHYA KARUNANITHI is currently an Assistant Professor Senior Grade-1 with the School of Computer Science Engineering (SCOPE), Vellore Institute of Technology, Vellore Campus, Tamil Nadu. She has been teaching for more than ten years and has guided undergraduate projects in the emerging trends. She organized and attended several FDPs, workshops, refresher courses, conferences, and technical talks in her domain and gained expertise. She also completed certification in the field of big data computing, data science, and machine learning. She has published several articles, patents, conference papers, journal articles, and authored book chapters. Her area of research interests include computer vision, artificial intelligence, machine learning, and big data analytics.

ARUNKUMAR BALAKRISHNAN received the Ph.D. degree in information science and engineering from Anna University, Chennai. He is currently an Associate Professor with the School of Computer Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India. He has 15 years of academic experience and ten years of parallel research experience in the field of cryptography, quantum medical processing, deep learning, LLM, and blockchain. His research interests include cryptography, network security, quantum medical image processing, blockchain, light-weight cryptography algorithms, and LLM.